AI-Based E-Commerce Product Representation Consistency and Reliability Assessment System

B.Tech CSE Major Project - Executive Concept & Architecture Guide

1. What is Our Project in One Sentence?

"An AI-based system that examines what an e-commerce seller claims about a product (material, sizing, color, functionality) and mathematically evaluates how consistently actual customer evidence (written reviews and uploaded photos) supports or contradicts those claims."

2. The Real-World Problem & Our Guiding Philosophy

When shopping on Amazon or Flipkart, consumers rely entirely on the digital representation: studio-lit photos and marketing bullet points ("100% Genuine Leather", "True to Size", "Navy Blue", "10-Hr Battery"). Often, the delivered product has synthetic fabric, runs two sizes too small, or fails within days.

Why existing platform metrics fail:
- Star Ratings are Misleading: A customer might write: 'Cute purse! 5 stars! But note: it is canvas, NOT real leather.' Standard sentiment analysis marks this 100% positive, ignoring the contradicted material claim.
- Fake Review Detectors Miss the Problem: Detecting bots is useful, but even with 500 authentic human reviews, no tool verifies if the seller's claims match what those humans experienced.

OUR PHILOSOPHY: Reliability Engineering, NOT 'Fraud Detection'
We never accuse sellers of criminal fraud or intentional lying (intent cannot be proven in code). Instead, we conduct an objective Quality Assurance audit: 'How consistently does available customer evidence support each specific claim?' If evidence conflicts, we report empirical contradiction percentages.

3. Project Foundation: Base Paper vs. Our Proposed Features

Our project is anchored in published top-tier research from ACM SIGIR 2022 by scientists from Amazon Israel and Meta:

Component	Referral Base Paper (Farber et al., SIGIR '22)	Our Proposed Extensions & Solution
Target Analyzed	Customer review tips (e.g., 'size up')	Official seller claims, specs & marketing bullet points
Data Modality	Purely text-based	Multimodal: Text reviews + Customer-uploaded photos
Visual Verification	None	CIE76 Delta E (LAB color space) + CLIP visual similarity
Output Metric	Support and Contradiction scores	Product Representation Index (PRI: 0-100%) + 4-Tier Status
Academic Link	Base citation anchor	Directly fulfills Section 5 (Future Work) of Farber et al.

4. How the System Works: 5 Simple Steps

Step 1: Claim Extraction (From Seller Description)

We deconstruct the seller's long paragraph into single, testable, atomic claims (Tp).
Example: 'Men's running shoes with breathable mesh and slip-resistant rubber sole. Fits true to size.'
-> T1: 'Upper is breathable mesh' [MATERIAL] | T2: 'Outsole is non-slip rubber' [FUNCTION] | T3: 'Fits true to size' [SIZING].

Step 2: Semantic Evidence Retrieval (Sentence-BERT)

A product has hundreds of reviews with thousands of chatter sentences ('fast delivery', 'nice box'). Sentence-BERT encodes sentences and retrieves only the Top-5 to 10 sentences specifically relevant to each claim.

Step 3: Text Natural Language Inference (RoBERTa NLI)

We test each review sentence against the seller claim using a RoBERTa Cross-Encoder:
- Premise: Review sentence ('Peeling after 2 days, pure plastic.')
- Hypothesis: Seller claim ('100% Genuine Leather')
- Model outputs exact mathematical probabilities: P(Support), P(Contradict), P(Neutral).

Step 4: Visual Discrepancy Engine (When Customer Photos Exist)

When customers post photos of what they received, we compare them against the official catalog photo:
- Dominant Color (CIE76 Delta E): Converts colors to CIE LAB space (human perceptual color space). Flags discrepancy if Delta E > 25.
- Visual Geometry (CLIP): Compares shape, pattern, and design using OpenAI's CLIP model.

Modularity Rule: If no customer photos exist, this step simply sleeps; the text engine runs 100% independently.

Step 5: Score Aggregation & Product Representation Index (PRI)

We mathematically aggregate support and contradiction scores across all claims and visual checks into a single 0-100% Product Representation Index (PRI) and assign a clear 4-tier status.

5. The 4 Final Assessment Outcomes (Taxonomy)

1. Consistent (High Reliability)
Customer evidence validates seller claims. Both specs and photos match what was delivered.

2. Contradictory (High Discrepancy)
Customer evidence refutes one or more core claims (e.g., synthetic fabric instead of silk, wrong color).

3. Mixed Evidence (Divided Feedback)
Customer experiences are split. Very common in sizing (some say 'fits true', others say 'size up').

4. Insufficient Evidence (Cold Start)
New product with 0 reviews or < 3 reviews. System flags unprovable marketing buzzwords instead.

6. Smart Edge-Case Handling Matrix

How our system deterministically handles complex real-world situations:

Scenario	System Detection & Logic	User Explanation & Outcome
0 Reviews (Cold Start)	No customer feedback available to test.	Insufficient Evidence: Runs intrinsic claim audit flagging unprovable superlativ
Sizing Controversy	High support AND high contradiction on fit.	Mixed Evidence: Informs buyer that 48% say true-to-size while 52% recomme
Dim / Warm Room Light	Customer photo has yellow tint; Delta E = 17, but CLIP consistent with visual ha	Consistent with Visual Note: System accounts for lighting variance without fal
High Rating, False Spec	5-star rating, but reviews state fabric is synthetic, not genuine leather.	Claim Contradicted: Flags specific material contradiction regardless of positive
Photo Matches, Dead Device	Customer photo looks identical, but reviews state 'battery died on day 2'.	Functional Failure: Visual appearance matches, but operational specific
No Customer Photos Uploaded	Reviews exist, zero customer photos.	Text Core Complete: Vision engine sleeps gracefully; zero system errors.

7. Technology Stack (Free & Open-Source)

- Sentence Embedding: sentence-transformers/all-MiniLM-L6-v2 (fast semantic sentence search)
- NLI Cross-Encoder: cross-encoder/nli-roberta-base (Support / Contradict / Neutral classification)
- Multimodal Vision: openai/clip-vit-base-patch32 (image visual feature similarity)
- Color Science: OpenCV & scikit-learn (RGB to CIE LAB conversion & K-Means clustering)
- Interactive UI: Streamlit web dashboard
- Benchmark Dataset: Amazon Review Data (2018) (UCSD / Julian McAuley)

8. Quick 30-Second Viva Defense Script

Q: Why not just use ChatGPT or Gemini API?
A: Commercial LLMs are non-deterministic, black-box, and prone to hallucination. An academic project requires a deterministic, reproducible mathematical framework. Our system runs locally using specialized open-source transformer models with formal mathematical aggregation based on Farber et al. (ACM SIGIR 2022).

Q: How is this different from Sentiment Analysis?
A: Sentiment analysis measures emotion (happy vs sad). Our system evaluates claim truthfulness. A customer can write a 5-star happy review while noting that a shirt is polyester rather than cotton. Sentiment analysis flags that as positive; our system detects that the seller's material claim was contradicted.